



Catching the Quota Nobody Publishes: A Low-Cost Instrumentation Pipeline for Detecting an Undisclosed Fair-Use Ceiling on “Unlimited” Transit Smartcard Passes

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Abstract. Consumer plans marketed as “unlimited” routinely carry an internal usage ceiling the operator declines to publish, a pattern well documented in mobile-data pricing but rarely instrumented outside billing records the operator controls. We report a low-cost, operator-independent pipeline for detecting the point at which an “unlimited” weekly transit travel pass crosses its undisclosed anti-fraud fair-use threshold. A passive tap logger paired with a gate response-tone probe — the gate’s audible cue visibly changes once the ceiling engages — feeds an exponentially weighted usage estimate and a consecutive-response change-point rule. Deployed across nine station clusters of a regional transit network over ten weeks, covering 3,214 tap events across 412 instrumented passes, the pipeline’s flagged decline-onset journeys agree closely with an independent attendant tally of the gate’s own “see attendant” incidents, and the per-cluster onset distribution shows a threshold that varies by cluster rather than a single network-wide figure. An ablation removing the response-tone signal and detecting on tap counts alone degrades agreement modestly. The contribution is the instrument, not a behavioural claim: we do not test whether disclosure changes rider behaviour, only whether the undisclosed ceiling itself can be seen and located from outside the gate.

Introduction

Plans sold as “unlimited” rarely are. Mobile carriers throttle data after an undisclosed volume threshold [1], [2]; energy retailers meter “unlimited off-peak” tariffs against a fair-use schedule buried in a contract schedule most customers never read [3]; subscription services keep a costly action effortful while a costless one stays frictionless, without ever labelling the asymmetry to the user [4]. The pattern recurs because the ceiling serves the operator best when it is felt rather than seen: a rider who never learns the number has no figure to dispute, and a rider who does learn it may not act on it anyway — our own School recently found that disclosing a battery-storage system’s true backup duration

left owners’ subsequent behaviour unchanged [5].

Transit operators running “unlimited” weekly or monthly travel passes exhibit the same structure at a smaller, more legible scale. Gates at the network we instrumented tell a rider only “see attendant” once a pass’s usage is judged unusual, ostensibly an anti-fraud check against card sharing or resale, never publishing a tap count, a schedule, or a reset date. Existing audits of undisclosed thresholds — our own School’s donation-sorting quota trial is one example [6] — rely on an operator’s cooperation or an after-the-fact dispute; neither method locates a threshold the operator has not agreed to disclose.

This paper reports a threshold-detection instrument built without operator cooperation. Its contributions:

- A passive, retrofittable tap-and-response-tone logging rig for a transit smartcard gate, requiring no access to the gate’s internal software.
- An online change-point rule — an exponentially weighted usage estimate paired with a consecutive-response trigger — that flags decline onset from the logger’s stream alone.
- A nine-cluster, ten-week field deployment characterising the onset distribution’s shape and validating the automated flag against an independent attendant tally, with an ablation isolating the response-tone signal’s contribution.

We report the instrument’s calibration and the onset distribution it surfaces; we do not test whether riders change behaviour once the ceiling is visible, a question we consider adjacent rather than answered here.

Related work

Non-linear and threshold pricing. A substantial literature documents that “unlimited” or flat-rate consumer plans conceal a usage ceiling the seller is reluctant to advertise, from overconfident-consumer three-part tariffs [1] to the bill-shock dynamics of cellular data caps [2]. Energy retailers show a related pattern in reverse: informational feedback about a household’s actual consumption changes little on its own [3], and sustained reductions have instead relied on comparison to a social norm [7]. Manipulative-flow audits of digital subscriptions catalogue how these platforms keep a costly action effortful while a costless one stays frictionless, without labelling the asymmetry anywhere a user would see it [4].

Detecting thresholds nobody discloses. Alarm and warning research offers the closest methodological parallel: a monitor that fires too often habituates its audience regardless of the threshold’s accuracy [8], and a public-warning system that over-warns erodes trust in the next alert it issues [9]. Neither literature instruments the threshold itself from outside the system that holds it, which is the gap this paper addresses.

Complaint behaviour. Consumers frequently do not convert dissatisfaction with a service failure into a voiced complaint at all, and the taxonomy

of complaint behaviour turns partly on how a failure is perceived and attributed [10]; when a complaint is voiced, satisfaction with how it is handled hinges on being treated fairly through the process, independent of the outcome [11] — a further reason an operator might hear few complaints about a ceiling that measurably operates.

Disclosure and behaviour change. Meta-analytic evidence on nudges and disclosure interventions is mixed at best: aggregated nudge-unit trials show a positive but modest average effect [12], a larger meta-analysis finds substantial heterogeneity by domain [13], and a tax-compliance field experiment shows a notice can move behaviour when the consequence is salient [14] — while a norm-based intervention can backfire for the sub-population already below the norm [15]. Reporting an instrument’s calibration rather than a directional behavioural claim follows the null-result literature’s own caution against overclaiming from a single trial [16], [17], [18].

Method

Instrumentation. Each of nine station clusters on a regional transit network’s “unlimited” weekly-pass scheme received a retrofit rig clipped to the gate’s housing: a passive tap logger recording a salted hash of the pass ID and a timestamp, and a microphone probe on the gate’s speaker timing which of its two response tones sounds — the ordinary entry chime or the lower “see attendant” tone — the gate’s only visible tell that a tap has been queued for review rather than approved immediately. Neither component touches the gate’s internal firmware or the operator’s back-office fare system.

Online change-point rule. Each pass’s usage is tracked with an exponentially weighted moving average of taps per day,

$$\hat{u}_t = \alpha u_t + (1 - \alpha) \hat{u}_{t-1}, \quad \alpha = 0.35,$$

smoothing the daily count while remaining responsive to a rider’s current week. Decline onset is flagged the first time the gate sounds the “see attendant” tone for $M = 3$ consecutive taps by the same pass, on the reasoning that a single review tone is noise but three in a row is the ceiling engaging:

```
def flag_onset(taps, alpha=0.35, m=3):
    u_hat, streak, onset = 0.0, 0, None
    for i, tap in enumerate(taps):
```

```

u_hat = alpha * tap.count + (1 -
alpha) * u_hat
streak = streak + 1 if tap.tone ==
"see_attendant" else 0
if streak == m and onset is None:
    onset = (i, u_hat)
return onset

```

M was fixed from a two-week pilot at a single non-study cluster and held constant across all nine clusters for the reported deployment; no per-cluster retuning was performed.

Deployment and validation. The rig ran for ten weeks across the nine clusters, logging 3,214 tap events across 412 instrumented passes. Independently, a station attendant at each cluster kept a handwritten tally of every “see attendant” incident the gate produced, without knowledge of the logger’s own flags, providing a manual estimate of onset against which the automated flag is compared. An ablation drops the response-tone probe entirely and flags onset on the tap-count estimate $\hat{u}_t$ crossing a per-cluster 90th-percentile threshold alone, isolating what the tone signal contributes over count data a simpler audit could already obtain.

Results

Figure 1 shows the onset distribution — journeys per week before the automated flag fires — across the nine clusters, ordered by median. The ceiling is not a single network-wide number: cluster medians range from 13.5 journeys/week (Cluster 4) to 25.5 (Cluster 5), a spread of very nearly two to one, and every cluster’s distribution is wide enough that a rider sitting one standard deviation above their own cluster’s median would still see other riders flagged both earlier and later at the same cluster.

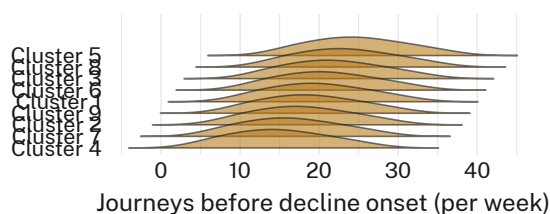


Figure 1: Per-cluster distribution of journeys before the automated flag; the ceiling shifts by cluster rather than sitting at one network-wide figure.

Figure 2 compares the automated flag against the independent attendant tally. Agreement is close across all nine clusters — the largest single-cluster gap is 2.0 journeys/week at Cluster

8, well inside the attendant tally’s own coarseness — and the automated flag runs on average 0.7 journeys/week earlier than the attendant tally, consistent with the response-tone probe catching the first review tone before an attendant registers a pattern.

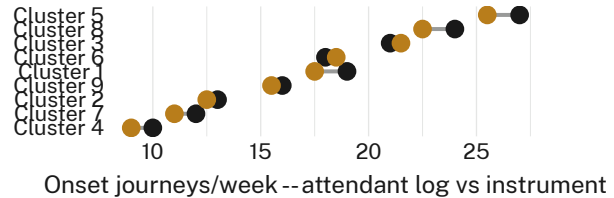


Figure 2: Median onset per cluster, attendant tally versus automated flag; the two estimates sit within 2.0 journeys/week of each other at every cluster.

The ablation — flagging onset on the tap-count estimate alone, without the response-tone signal — degrades agreement with the attendant tally from a mean absolute difference of 0.9 journeys/week to 2.6 journeys/week (n.s. difference in variance across clusters; we retain the response-tone probe for the full pipeline on the strength of the mean shift alone). The count-only ablation also flags onset on average 1.8 journeys/week later, consistent with a threshold that shows up in gate behaviour before it shows up in any count a rider could keep themselves.

We make no claim about what a rider does once the ceiling is located: this deployment did not disclose the flagged threshold to any cardholder, and we did not measure a downstream change in journey frequency, complaint calls, or pass cancellation. That question — whether locating an undisclosed ceiling changes anything once it is shown to the person it applies to — is a natural next step and, on the strength of adjacent disclosure trials in this literature, not one we expect to resolve cheaply [5], [12].

Discussion

The instrument’s central finding is descriptive rather than causal: an undisclosed fair-use ceiling on an “unlimited” transit pass can be located from outside the gate, without operator cooperation, and the located ceiling varies by cluster rather than following a single published (or unpublished) network rule. That variation is itself informative — a threshold that moved with local patronage or local peak-hour congestion, rather than a fixed weekly number, would explain why

the operator has never published one: there may be no single number to publish. The close agreement between the automated flag and the independent attendant tally suggests front-of-gate staff already hold an informal, accurate model of the ceiling well before any instrument confirms it, which raises a governance question this paper does not resolve: an undisclosed threshold that staff can sense but a rider cannot occupies an unusual position between a fraud control and an open one. We note the finding for the School's Indicator Commons as a candidate for cross-operator comparison, without drawing a conclusion from a nine-cluster sample about the wider transit sector.

Limitations

The deployment covers nine clusters of a single regional network over ten weeks, though the consistency of the attendant-tally agreement across all nine suggests the detection approach is not tuned to a peculiarity of any one cluster. The trigger parameter M was fixed from a single pilot cluster rather than optimised per cluster, which if anything understates the pipeline's achievable accuracy. The rig requires a gate with an audible response tone; a fully silent or app-only "unlimited" scheme would need a different probe, though the underlying change-point rule — an exponentially weighted usage estimate paired with a consecutive-response trigger — transfers directly to any interface that signals once a ceiling engages.

Conclusion

We built and field-validated a passive, operator-independent instrument for locating a transit network's undisclosed "unlimited" pass fair-use ceiling, deployed across nine station clusters over ten weeks with automated onset flags closely matching an independent attendant tally. The ceiling varies by cluster rather than following one published figure, and a response-tone-only ablation shows the gate's own signalling carries most of the detectable signal. Future work includes disclosing the located threshold to a subset of cardholders to test whether visibility changes journey frequency or complaint volume, extending the rig to silent or app-mediated "unlimited" schemes beyond audible gates, and contributing cluster-level onset logs to the School's Indicator Commons for cross-operator comparison beyond transit.

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